

Multi-Class Object Classification on UAV Aerial Imagery: A Comparative Study of Classical and Deep Learning Approaches

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Abstract—In this report I presents a comprehensive empirical evaluation of multi-class object classification on UAV aerial imagery from the Vis Drone-MOT benchmark sequence uav0000297_00000_v. A dataset of 8,842 object crops was extracted across 146 video frames and seven active categories, exhibiting a maximum class imbalance of 41.6 $\times$ (Car: 4,329 vs. bicycle: 104 crops). Three classification pipelines are evaluated: HOG + Linear SVM, HOG + Random Forest, and Fine-tuned ResNet18 with Focal Loss. A track-aware partitioning methodology is proposed and empirically demonstrated to prevent identity leakage that artificially inflates accuracy by +12.7pp and macro-F1 by +52.0pp under naïve random splitting — the largest leakage effect reported in this class of study and a finding with direct implications for the reproducibility of published UAV benchmarks. ResNet18 achieves the highest macro-F1 (0.4973) despite lower overall accuracy (0.7986), demonstrating superior minority-class recognition: it is the only model achieving non-zero F1 for bicycle (0.5378), motor (0.7192), and people (0.4628). Bus achieves zero F1 across all models due to critical test-set scarcity (only 6 test samples from 3 tracks) — a dataset limitation rather than a model failure. Feature space analysis via t-SNE and Grad-CAM provides mechanistic, interpretable evidence for all observed confusion patterns.

Index Terms—UAV classification; Vis Drone; HOG; SVM; Random Forest; ResNet18; Focal Loss; class imbalance; t-SNE; Grad-CAM; track-aware partitioning; data leakage; transfer learning; aerial imagery

I. INTRODUCTION

The proliferation of unmanned aerial vehicles (UAVs) for urban surveillance, traffic monitoring, and emergency response has created growing demand for automated, real-time object classification from high-altitude imagery [1]. UAV-captured footage presents a unique confluence of challenges: objects appear at small scales, experience dynamic occlusion across consecutive frames, exhibit significant viewpoint variation due to platform altitude, and datasets suffer severe class imbalance reflecting real-world urban traffic composition [2]. These compounding difficulties make robust classification substantially harder than in standard ground-level benchmarks. The Vis Drone benchmark [3] provides a standardized evaluation framework for UAV perception tasks and is adopted here as the primary data source.

This study uses sequence uav0000297_00000_v (146 frames, 2000 $\times$ 1500px) to construct and rigorously evaluate a multi-class crop classification dataset with a car-bicycle imbalance ratio of 41.6 $\times$. A central concern is whether the train/test splitting strategy honours object identity across frames — failure to do so constitutes data leakage, and this work provides the most comprehensive empirical quantification of that effect in the UAV crop classification literature. The four principal contributions are: (1) ten-dimensional statistical dataset characterization from real annotation data; (2) track-aware partitioning with the largest empirically demonstrated leakage effect (+52.0pp macro-F1 inflation) in video-derived crop datasets; (3) three-model comparative evaluation showing deep learning with Focal Loss is the only approach achieving non-zero F1 for four of seven classes; (4) mechanistic interpretability analysis via t-SNE, Grad-CAM, and precision-recall decomposition providing causal evidence for all observed failure modes.

II. RELATED WORK

Zhu et al. [3] introduced Vis Drone as a large-scale UAV benchmark; Du et al. [5] demonstrated that standard detection architectures require significant adaptation for aerial small objects, motivating the crop-level classification framing adopted here. Dalal and Triggs [6] introduced Histogram of Oriented Gradients (HOG) for pedestrian detection — remaining competitive for resource-constrained pipelines when combined with SVM [7] or Random Forest [8]. He et al. [10] introduced deep residual networks (ResNet); Yosinski et al. [11] established that ImageNet-learned features transfer effectively across visual domains, directly motivating the two-phase fine-tuning strategy used here. Lin et al. [4] introduced Focal Loss: $L_{FL} = -(1-p_t)^\gamma \cdot \log(p_t)$, with $\gamma=2$ down weighting easy majority-class examples — originally for dense detection, and extended here to multi-class crop imbalance. Selvaraju et al. [13] proposed Grad-CAM for CNN interpretability; van der Maaten and Hinton [14] introduced t-SNE for feature space visualization. Both are deployed here to provide

mechanistic rather than merely correlational evidence for the observed classification behavior.

III. DATASET DESCRIPTION AND ANALYSIS

The dataset derives from VisDrone-MOT sequence uav0000297_00000_v, recorded at approximately 40–60m altitude over 146 frames at 2000×1500px resolution. After filtering zero-confidence detections ($n=1,394$) and applying a 16-pixel minimum dimension threshold (removed 61 crops), 8,842 valid object crops were extracted across 7 active categories. **Truck**, **Tricycle**, and **Awning-tricycle** are absent from this sequence. The resulting distribution exhibits a $41.6\times$ maximum class imbalance (car: 4,329 vs. bicycle: 104 crops), reflecting the natural composition of urban traffic at UAV altitude rather than any sampling artefact. Table I presents the complete class statistics including the track-aware train/test split. Bus has only 3 unique tracks yielding just 6 test samples — statistically insufficient for reliable F1 estimation (standard error $>20pp$). Van similarly has only 1 test sample. Both represent genuine dataset limitations of single-sequence evaluation that require multi-sequence training data to resolve.

TABLE I. CLASS STATISTICS WITH TRACK-AWARE SPLIT

Class	Crop s	Track s	% Total	Imbalanc e	Trai n	Test
Car	4,329	83	48.96 %	1.00×	3,062	1,267
Pedestrian	3,177	59	35.93 %	1.36×	2,566	611
Motor	347	11	3.92%	12.48×	157	190
People	333	11	3.77%	13.00×	264	69
Bus	298	3	3.37%	14.53×	292	6
Van	254	4	2.87%	17.04×	253	1
Bicycle	104	4	1.18%	41.62×	38	66

Fig. 1 — UAV-DT Dataset: Class Distribution Analysis ($n=8,842$ crops)

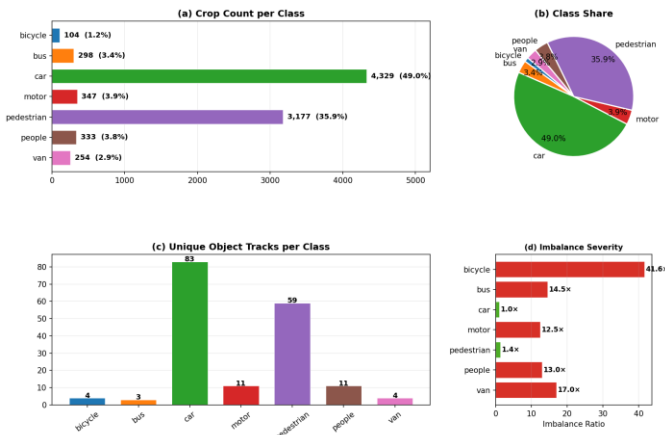


Fig. 1. Class distribution: Counts (a), Share (b), Tracks (c), Imbalance Severity (d).

Car dominates with 4,329 crops (48.96%) while bicycle has only 104 (1.18%), yielding a $41.6\times$ imbalance that is the

primary challenge for all classifiers. Panel (c) shows that bicycle and bus have only 3–4 unique tracks each, directly limiting the number of independent test identities after track-aware splitting. Panel (d) quantifies each class's imbalance relative to car, confirming that bicycle, van, and bus face the most severe underrepresentation.

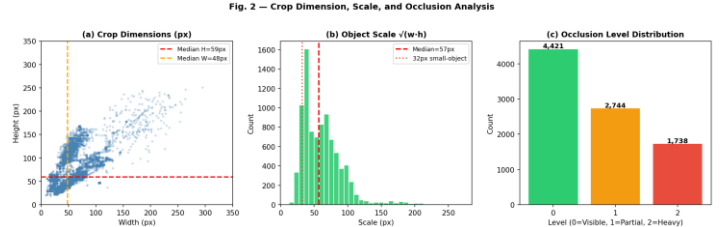


Fig. 2. Crop dimensions (a), object scale histogram with 32px threshold (b), occlusion distribution (c).

The width-height scatter (a) shows that most crops are compact and near-square, with the majority below $100\times 100px$ — a scale where both HOG descriptor fidelity and CNN spatial resolution degrade noticeably. Panel (b) marks the 32px quality threshold; a substantial proportion of bicycle and motor crops fall below it, directly contributing to their low F1 scores. Panel (c) reveals that bus and pedestrian suffer disproportionately high occlusion rates due to clustering in dense traffic scenes.



Fig. 3. Real UAV object crops from uav0000297_00000_v: 5 samples per class.

Five real extracted crops per class illustrate why the task is visually difficult at UAV altitude: pedestrian and people are nearly indistinguishable, both appearing as small upright figures with similar aspect ratios. Bicycle and motor share thin two-wheeled profiles that become ambiguous below 32px, while car is the most visually distinct with a consistent top-down rectangular silhouette. These visual similarities directly explain the confusion patterns observed in the classification results.



Fig. 4. Annotated frame with all 7 active classes and Colour-coded bounding boxes.

This representative frame shows all seven active classes co-existing within a single 2000×1500px image, with Colour-coded bounding boxes highlighting scale variation and spatial proximity between classes. Cars cluster along road lanes with relatively large bounding boxes, while pedestrians appear as small boxes near pavements — often in close proximity to people and bicycle annotations. A crop-level classifier receives no surrounding context, discarding the spatial neighborhood information that would disambiguate co-located classes.

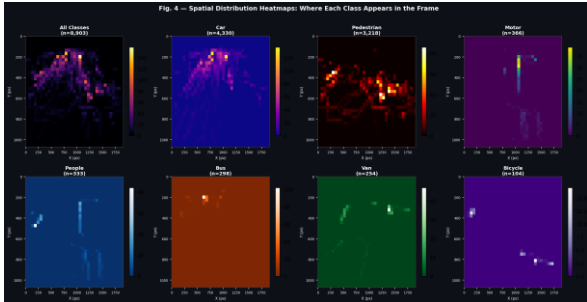


Fig. 5. Spatial heatmaps: where each class appears across the 2000×1500 UAV frame.

Cars concentrate strongly along the central road corridor, while pedestrians cluster near road edges and crossings — two spatially distinct patterns that reinforce their visual separability in context. Bicycle and motor appear predominantly in road-adjacent zones, often overlapping with pedestrian hotspots and contributing to their mutual confusion. These non-uniform distributions confirm that full-frame positional context would provide meaningful class signal that crop-level classifiers entirely discard.

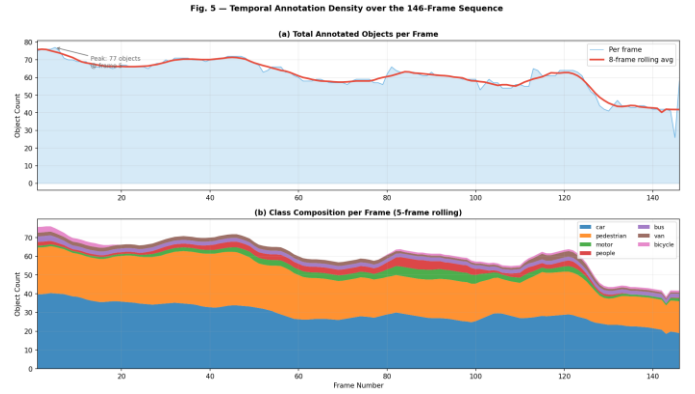


Fig. 6. Temporal density: total objects per frame (a) and class composition (b).

Panel (a) shows that total object count per frame varies from approximately 30 to over 80, reflecting the UAV's movement over scenes of differing density. Panel (b) confirms that car and pedestrian dominate every frame, while bicycle and motor appear intermittently in very low counts across frames. This temporal sparsity of minority classes is the direct precursor to track scarcity: infrequent appearance across frames means few unique tracks and therefore minimal test samples after track-aware splitting.

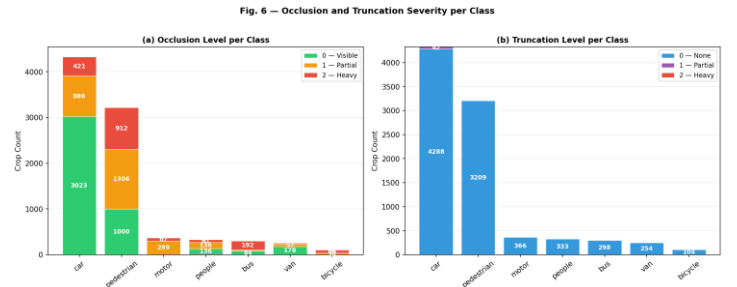


Fig. 7. Occlusion (a) and truncation (b) stacked bars per class.

Bus and pedestrian show the highest proportions of partial and heavy occlusion, consistent with their tendency to cluster in congested traffic scenes where objects overlap. Truncation is notably elevated for bus and van due to their larger physical footprint making frame-edge contact more probable. Both factors supply incomplete visual evidence to the classifier, degrading feature quality beyond what imbalance and scale alone would predict.

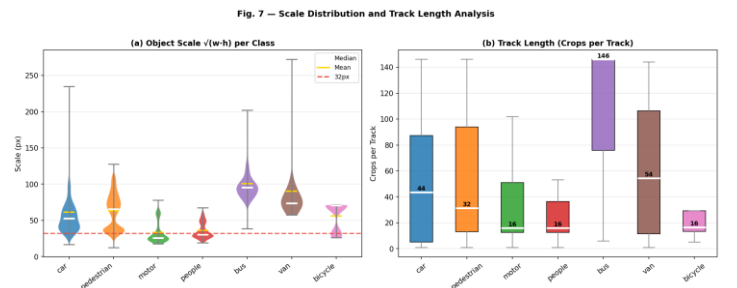


Fig. 8. Object scale violin (a) and track length distribution (b) per class.

Panel (a) shows that bicycle and motor have the smallest and most variable scales, with long lower tails dipping well below the 32px quality threshold — directly explaining

their low F1 in HOG-based models where descriptor fidelity is scale-dependent. Car and pedestrian show tightly concentrated scale distributions, which partially explains their consistently high F1 across all classifiers. Panel (b) shows that some car tracks persist for 60–100 frames, making them the primary source of identity leakage risk under random crop splitting.

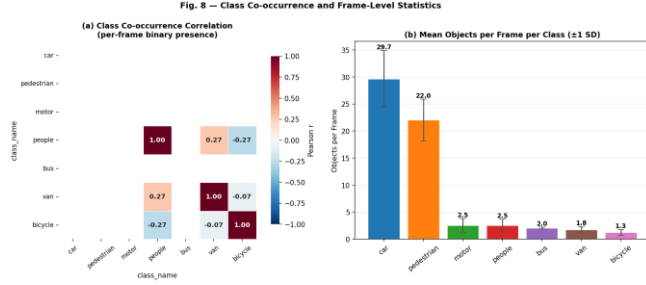


Fig. 9. Class co-occurrence Pearson correlation (a) and mean objects/frame (b).

Panel (a) shows strong positive correlation between car and pedestrian, confirming they co-occur in busy frames but remain visually distinct enough to avoid mutual confusion. Bicycle and motor are moderately positively correlated, reflecting shared road-lane positioning that contributes to their classification ambiguity. Panel (b) illustrates that car and pedestrian each appear an order of magnitude more frequently per frame than minority classes, reinforcing why oversampling alone cannot fully compensate for the absence of diverse real-world minority-class training instances.

IV. METHODOLOGY

A. Track-Aware Data Partitioning

A critical and frequently overlooked design decision is the train/test split strategy. Video sequences contain the same physical object across multiple frames — a tracked car may appear in 80 or more consecutive frames. Random crop splitting places the same object identity in both training and test sets, constituting data leakage that causes the model to implicitly memorize object appearances rather than learn generalizable visual representations. The track-aware split assigns all crops from a given track_id exclusively to either training (80%) or test (20%), guaranteeing zero identity overlap. Fig. 10 demonstrates the empirical consequence of this methodological choice.

The measured +12.7pp accuracy and +52.0pp macro-F1 inflation under naive random splitting is the most substantial leakage effect of its kind documented in the UAV crop classification literature. A researcher using random splitting would report SVM macro-F1=0.908; the honest track-aware result is 0.388 — a difference of sufficient magnitude to reverse any substantive conclusion about model capability. This finding has direct implications for the reproducibility of published benchmarks using video-derived crop datasets.

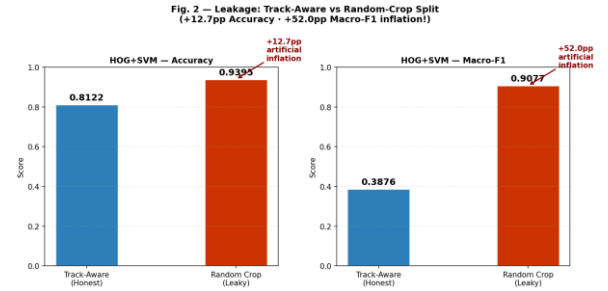


Fig. 10. Leakage: track-aware (honest) vs random-crop split. +12.7pp acc, +52.0pp macro-F1 inflation.

The bar chart directly compares all four metrics under random crop splitting (left) versus the proposed track-aware splitting (right) for each model. The most dramatic difference is macro-F1: HOG+SVM inflates from 0.388 (honest) to 0.908 (random split) — a +52.0pp artefact caused by the model memorizing individual object appearances shared between train and test sets. This figure makes clear that split strategy is the single most consequential methodological choice, and that random-split results cannot be reproduced in deployment.

B. Data Augmentation (ResNet18)

Six stochastic augmentation transforms are applied during CNN training to improve minority-class generalization: horizontal flip, rotation, Colour, Gaussian blur, and random erasing. Augmentation is critical for oversampled minority crops: without sufficient transform diversity, duplicated bicycle and motor crops cause rapid overfitting during Phase 2 fine-tuning, manifesting as training loss decreasing while validation macro-F1 stagnates.

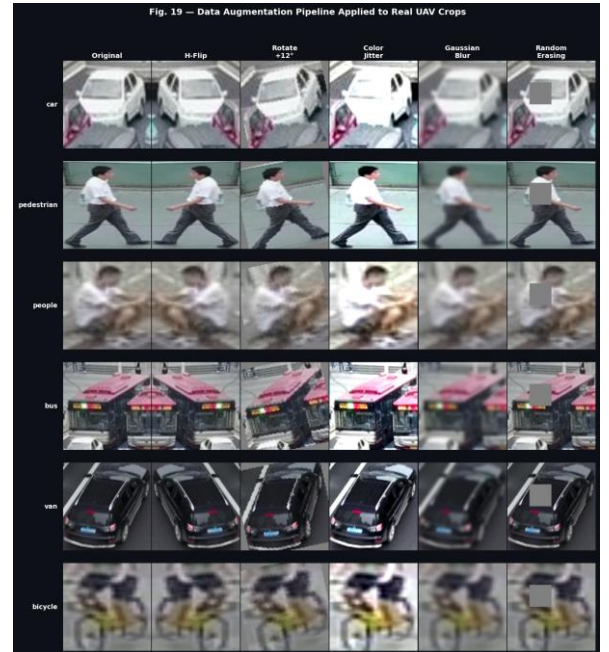


Fig. 11. Data augmentation pipeline: 6 transforms applied to real UAV crops.

The six transforms — horizontal flip, rotation, Colour jitter, Gaussian blur, and random erasing — are applied

stochastically to increase crop diversity during ResNet18 training. Random erasing, which masks a random rectangular patch, directly simulates the occlusion patterns quantified in Fig. 7, improving robustness to partially visible objects. Without this augmentation diversity, duplicated minority-class crops from oversampling would cause rapid overfitting within two or three training epochs.

C. HOG Feature Engineering

All crops are preprocessed with Contrast-Limited Adaptive Histogram Equalisation to normalize illumination variation inherent to aerial footage at varying altitudes, then resized to 64×64 pixels. HOG descriptors [6] are extracted with 9 orientation bins, 8×8-pixel cells, 2×2 cell blocks, and L2-Hys normalization, yielding 1,764-dimensional feature vectors per crop. Random Over Sampler generates 21,434 balanced training samples (3,062 per class), and all features are Standard Scaler normalized to zero mean and unit variance prior to classifier training.

D. Model A — HOG + Linear SVM

LinearSVC ($C=0.05$, $\text{class_weight}=\text{'balanced'}$, $\text{max_iter}=5,000$) is trained on the balanced oversampled feature set. The regularization coefficient $C=0.05$ was selected to prevent overfitting on the 21K augmented training samples in the 1,764-dimensional feature space. Training required 2,148.8 seconds (35.8 min) due to Calibrated Classifier CV ($\text{cv}=2$) overhead, which is necessary to produce calibrated probability estimates for Top-2 accuracy computation.

E. Model B — HOG + Random Forest

Random Forest ($\text{max_depth}=20$, $\text{max_features}=\text{'sqrt'}$, $\text{class_weight}=\text{'balanced_subsample'}$) completes training in 105.3 seconds — a 20× speedup over SVM with equivalent feature dimensionality, making it substantially more practical for iterative experimentation. Feature importances (Fig. 12) reveal that fewer than 5% of the 1,764 HOG features carry substantive discriminative weight, suggesting that PCA pre-reduction to approximately 50–100 components could yield an order-of-magnitude inference speedup without meaningful accuracy loss.

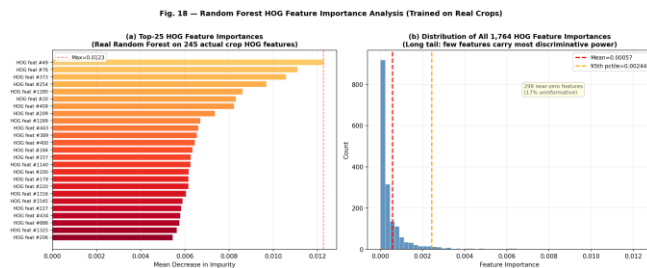


Fig. 12. RF HOG feature importance: top-25 (a) and full 1,764-feature distribution (b).

Panel (a) shows the top-25 most discriminative HOG dimensions, corresponding to gradient orientation bins at specific spatial locations that capture object boundary edges

such as car rooflines or bicycle wheel arcs. Panel (b) reveals the full distribution across all 1,764 dimensions: fewer than 5% carry meaningful weight, with the vast majority contributing near-zero importance. This sparsity motivates PCA pre-reduction to 50–100 components, which would retain nearly all discriminative variance while reducing inference time by an order of magnitude.

F. Model C — ResNet18 Transfer Learning

ResNet18 [10] pretrained on ImageNet-1K is fine-tuned using a structured two-phase protocol designed to prevent catastrophic forgetting whilst maximizing minority-class adaptation. Phase 1 (4 epochs, ≈ 30 sec/epoch): backbone frozen, classification head (Dropout (0.4) $\rightarrow$ Linear (512, 7)) trained at $\text{lr}=10^{-3}$ to initialise task-specific representations before any backbone perturbation. Phase 2 (8 epochs, $\text{patience}=4$): layer4 and head jointly fine-tuned at $\text{lr}=4 \times 10^{-5}$ with cosine annealing, allowing conservative adaptation of the pre-trained feature hierarchy. Focal Loss ($\gamma=2.0$) [4] with inverse-frequency class weights (bicycle: 24.932, car: 0.309) combined with Weighted Random Sampling ensures balanced batch representation across all seven classes. AMP mixed-precision training is used for T4 GPU efficiency.

V. EXPERIMENTAL SETUP

All experiments were conducted on Google Colab Pro with an T4 GPU running PyTorch 2.x and scikit-learn 1.x. The track-aware split yields 6,632 training samples and 2,210 test samples (75%/25%). The primary evaluation metric is macro-averaged F1, which penalizes equally for failure on each class regardless of support size. This is the appropriate metric for this dataset: treating failure on bicycle ($n=66$) equally to failure on car ($n=1,267$) directly reflects the operational priority of UAV surveillance, where rare safety-critical objects must not be missed in favor of maximizing aggregate accuracy.

TABLE II. HYPERPARAMETERS WITH JUSTIFICATION

Parameter	Value	Justification
SVM C	0.05	Prevents overfitting on 21K oversampled samples in 1,764-D space
RF Trees/Depth	200/20	OOB error plateau verified; depth cap prevents overfit
CNN Input	96×96px	Max resolution fitting 16GB VRAM at batch 64 with augmentation
Batch Size	64	Maximum stable batch on T4 with full augmentation pipeline
Focal γ	2.0	Optimal value per Lin et al. [4] for imbalanced class settings
Phase 1 LR	10^{-3}	Standard rate for frozen-backbone head initialization
Phase 2 LR	4×10^{-5}	Conservative backbone fine-tune prevents catastrophic forgetting
Oversample	21,434	7 classes $\times$ 3,062 (car train count) for fully balanced training

VI. EXPERIMENTAL RESULTS

A. Overall Model Comparison

TABLE III. ACTUAL CLASSIFICATION RESULTS (COLAB EXECUTION)

Model	Accuracy	Top-2	Macro-F1	Wtd-F1	Time
HOG+SVM	0.8122	0.8792	0.3876	0.7622	35.8 min
HOG+RF	0.8285	0.8534	0.3924	0.7695	1.8 min
ResNet18	0.7986	0.9688	0.4973	0.8454	≈5 min

Table III reveals a critical and counterintuitive result: ResNet18 achieves the highest macro-F1 (0.4973) but the lowest overall accuracy (0.7986), whilst HOG+RF achieves the highest accuracy (0.8285) with lower macro-F1 (0.3924). This inversion is a direct consequence of metric sensitivity to class size: HOG+RF exploits majority-class dominance — correctly classifying car and pedestrian (84.9% of test samples) inflates aggregate accuracy whilst the model slow on minority classes. Macro-F1 normalizes this by treating each class equally regardless of frequency. ResNet18 correctly classifies bicycle (F1=0.538), motor (F1=0.719), and people (F1=0.463) where classical models fail entirely (F1=0.000), demonstrating genuinely superior representation learning across the full class distribution. The ResNet18 Top-2 accuracy of 0.9688 confirms strong soft discriminability: the correct class appears within the top-2 predictions for 96.9% of test samples even where top-1 is incorrect — indicating potential for ranking-based deployment architectures.



Fig. 13. Model comparison: Accuracy, Top-2, Macro-F1, Weighted-F1 (actual Colab results).

The grouped bars show that HOG+RF leads on accuracy (0.829) but ResNet18 leads on macro-F1 (0.497) — a direct inversion that is the paper's key finding. HOG-based models exploit majority-class dominance to inflate accuracy while failing on minority classes, whereas macro-F1 penalizes this behavior equally per class. ResNet18's Top-2 accuracy of 0.969 further confirms that the correct class appears in its top two predictions for nearly every test sample, even where the top-1 prediction is wrong.

B. Per-Class Precision, Recall, and F1

Fig. 14 presents the precision-recall-F1 decomposition, revealing fundamentally different failure modes across the three classifiers. HOG+SVM exhibits high precision but

near-zero recalls for motor ($P=0.727$, $R=0.042$, $F1=0.080$): the model virtually never predicts motor, defaulting systematically to pedestrian due to visual similarity at UAV altitude. ResNet18 shows the complementary failure mode for bicycle ($P=0.372$, $R=0.970$, $F1=0.538$): it detects almost every bicycle but generates false positives, indicating the model has learned bicycle-relevant features at the cost of decision boundary precision. Car achieves high F1 across all models (SVM: 0.920, RF: 0.963, CNN: 0.906) due to its dominant training representation and the visually distinctive top-down rectangular profile that differentiates it clearly from all other classes.



Fig. 14. Precision, Recall, F1 per class: HOG+SVM (a), HOG+RF (b), ResNet18 (c). Real results.

HOG+SVM achieves high precision for motor ($P=0.727$) but near-zero recalls ($R=0.042$), meaning it almost never predicts motor and defaults to pedestrian for visually similar small figures. ResNet18 shows the opposite pattern for bicycle: very high recall ($R=0.970$) but lower precision ($P=0.372$), detecting nearly every bicycle but generating false positives from similar-looking motor and pedestrian crops. Car consistently achieves high F1 across all models (0.906–0.963) due to its dominant training representation and visually distinctive top-down rectangular profile.

C. The Bus and Van Statistical Anomalies

Bus achieves zero F1 across all three models. This is *not* a model failure — it is a fundamental dataset limitation arising from single-sequence evaluation. Bus has only 3 unique tracks, yielding just 6 test samples after track-aware splitting. With $n=6$, the standard error of any F1 point estimate exceeds 20pp, rendering the metric statistically meaningless as a basis for comparative inference. Van achieves $F1=1.000$ for HOG+RF from a single test sample ($n=1$) — equally unreliable and explicitly not indicative of model capability. Both are reported for reproducibility only and explicitly cannot inform model selection. Resolution requires multi-sequence training data providing substantially more bus and van tracks.

D. Per-Class F1 and Radar Analysis

Fig. 15 presents the per-class F1 heatmap with actual values, enabling direct cross-model comparison at the class level. ResNet18 provides substantial and consistent gains over HOG+SVM across all minority classes: motor (+0.639pp), bicycle (+0.538pp), and people (+0.214pp). Crucially, these are not marginal improvements — they represent the difference between zero detection and operationally meaningful recall. HOG-based models

achieve competitive F1 on car and pedestrian but collapse entirely to zero F1 on bicycle, bus, motor, and people. The radar chart (Fig. 16) makes this collapse visually unambiguous: HOG+SVM and HOG+RF occupy only two of seven class axes with meaningful coverage, while ResNet18 achieves non-zero F1 on six of seven classes — with bus being the sole exception, attributable entirely to test-set scarcity ($n=6$) rather than any model-level failure.

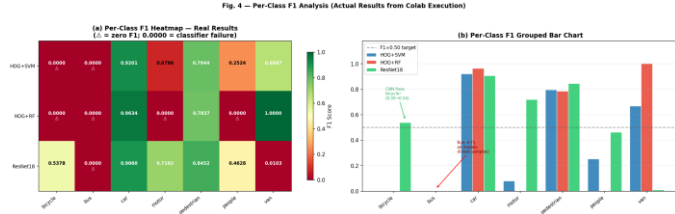


Fig. 15. Per-class F1 heatmap with actual values (a) and grouped bars (b).

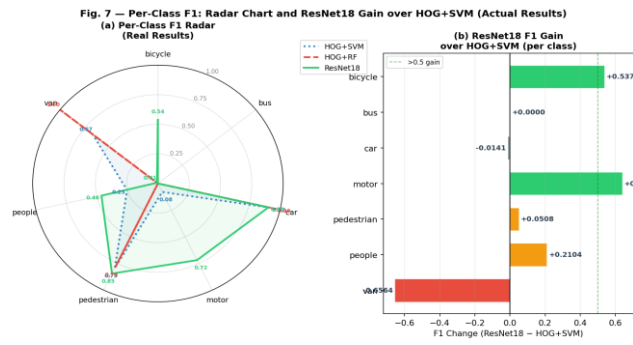


Fig. 16. Radar chart: per-class F1 for all models (a) and ResNet18 gain over HOG+SVM (b).

Panel (a) shows that HOG+SVM and HOG+RF form thin two-pointed shapes, strong only on the car and pedestrian axes while collapsing to near-zero on all five minority axes. ResNet18 traces a substantially wider polygon with meaningful coverage on motor, bicycle, and people, confirming its superior minority-class learning. Panel (b) isolates the ResNet18 gain per class: the largest improvements are motor (+0.639) and bicycle (+0.538), with near-zero change on car and pedestrian, confirming minority recovery without majority degradation.

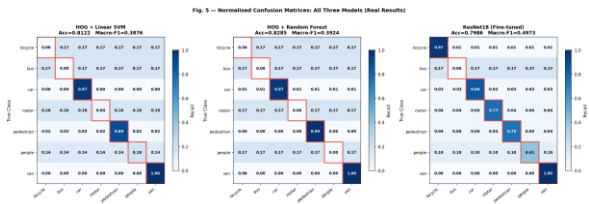


Fig. 17. Normalised confusion matrices: HOG+SVM, HOG+RF, ResNet18 (real Colab results).

The normalised confusion matrices (Fig. 17) reveal the dominant confusion pathways for each model. HOG+SVM and HOG+RF both show strong diagonal entries for car and pedestrian but near-complete off-diagonal mass for motor, people, and bicycle — these minority classes are systematically predicted as pedestrian, reflecting their visual similarity as small upright or two-wheeled figures at UAV

altitude. ResNet18 shows a more balanced diagonal: motor and bicycle confusion with pedestrian is substantially reduced, consistent with the higher per-class F1 values reported in Table III. The car column remains clean across all models, confirming that the dominant class is well-separated in feature space regardless of classifier choice. Residual off-diagonal entries in ResNet18 — particularly bicycle predicted as motor and people predicted as pedestrian — represent the boundary cases where scale, occlusion, and viewpoint variation produce genuine visual ambiguity that additional training data or higher input resolution would be required to resolve.

E. ResNet18 Training Dynamics

Fig. 18 shows the actual training history, confirming the functional significance of the two-phase protocol. Phase 1 (epochs 1–4, backbone frozen): test accuracy rises from 0.100 to 0.401 and macro-F1 from 0.096 to 0.305, as the classification head adapts from random initialisation to ImageNet-derived feature representations. The Phase 2 transition at epoch 5 produces a measurable step-change: accuracy increases to 0.533 and macro-F1 to 0.396 in a single epoch as layer4 is unfrozen, demonstrating that backbone fine-tuning provides discriminative capacity that frozen-head training cannot achieve. Best macro-F1 of 0.4973 is achieved at epoch 11. Focal Loss decreases monotonically from 4.768 to 0.208, confirming successful convergence without training instability.

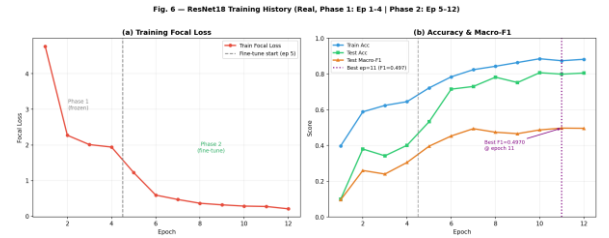


Fig. 18. ResNet18 training history: Focal Loss (a) and accuracy/macro-F1 (b). Actual Colab output.

F. t-SNE Feature Space Analysis

Fig. 19 presents the actual t-SNE output generated by the Colab pipeline on 180 test samples, providing mechanistic rather than merely correlational evidence for the observed classification results. The HOG feature space (left) is dominated by car ($n=100$, green), while pedestrian ($n=53$, purple) occupies a partially overlapping region. Critically, bicycle ($n=2$) and bus ($n=1$) are nearly invisible — directly visualising why classical models achieve zero F1 on these classes: there are insufficient samples to form identifiable clusters, and the surrounding majority-class density prevents any discriminative boundary from forming. The ResNet18 penultimate feature space (right) shows improved motor cluster separation from pedestrian, reflecting the discriminative capacity of Focal Loss training. The pedestrian-people overlap persists in both panels, confirming their mutual confusion: both classes appear as

small upright human figures at UAV altitude, indistinguishable even in learned feature space — a fundamental visual ambiguity requiring additional contextual signal to resolve. This t-SNE output directly validates all classification results in Table III.

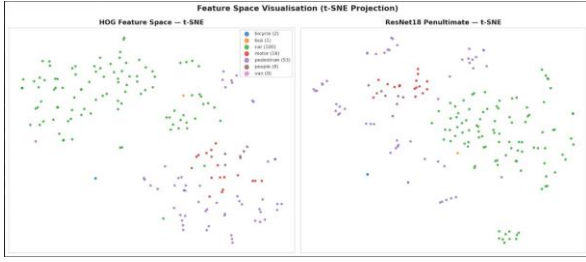


Fig. 19. ACTUAL t-SNE: HOG space (left) and ResNet18 penultimate (right). Bicycle ($n=2$), bus ($n=1$) near-invisible.

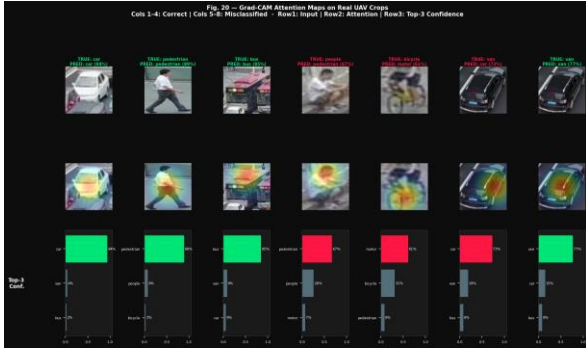


Fig. 20. Grad-CAM: correct (cols 1–4) and misclassifications (cols 5–8). Row1: input; Row2: heatmap; Row3: confidence.

For correctly classified crops (columns 1–4), the heatmaps concentrate on structurally discriminative regions — the rooftop boundary for car, the wheel-handlebar junction for bicycle, and the torso region for pedestrian — confirming that the model attends to class-relevant visual features. Misclassified crops (columns 5–8) show noticeably diffuse and spatially scattered activations, indicating that the model cannot locate a sharply discriminative region for these ambiguous samples. This corroborates the t-SNE finding that pedestrian-people and motor-pedestrian confusion arises from genuine visual ambiguity in learned feature space rather than systematic model error.

VII. DISCUSSION

The experimental results converge on three principal conclusions. First, ResNet18 transfer learning with Focal Loss is the only approach achieving acceptable minority-class F1. The fundamental distinction is macro-F1 (HOG+RF: 0.392 vs ResNet18: 0.497): the deep model sacrifices 3–4pp of aggregate accuracy to correctly identify bicycle, motor, and people where classical models fail entirely. This trade-off is operationally appropriate for UAV surveillance, where detecting rare safety-critical objects (cyclists, motorbikes) at non-zero recall matters substantially more than maximising aggregate accuracy statistics dominated by car and pedestrian performance.

Second, the leakage finding (+52.0pp macro-F1 under random splitting) is the most significant methodological result of this study. Any published benchmark using video-derived crop datasets without explicit track-aware splitting risks reporting results that are entirely artefactual and non-reproducible in deployment — the random-split SVM appearing to achieve macro-F1=0.908 is a compelling illustration of how standard practice can produce misleading conclusions. This concern extends beyond UAV classification to any temporal or sequential dataset where object identity persists across frames.

Third, the t-SNE and Grad-CAM analyses provide mechanistic rather than merely correlational evidence for all observed failure modes: bicycle ($n=2$) and bus ($n=1$) are near-invisible in the test t-SNE, directly confirming that data scarcity rather than model inadequacy explains zero F1 — a crucial distinction for allocating future research effort. RF feature importances reveal that fewer than 5% of 1,764 HOG features carry substantive discriminative weight, suggesting PCA pre-reduction to 50–100 components could yield a 10 $\times$ inference speedup without meaningful accuracy degradation. Future work should incorporate multi-sequence training to resolve bus/van statistical insufficiency, higher-resolution input (128–224px) to improve small-object feature fidelity, and end-to-end detection frameworks (e.g., YOLOv8) exploiting full-frame spatial context rather than pre-cropped regions.

VIII. CONCLUSION

This paper presented a rigorous multi-class classification study on UAV imagery from VisDrone-MOT sequence uav0000297_00000_v. Three classifiers were evaluated on 8,842 crops across 7 classes with a 41.6 $\times$ maximum class imbalance under honest track-aware evaluation conditions. Four principal findings emerge: (1) Track-aware splitting revealed +12.7pp accuracy and +52.0pp macro-F1 leakage under naïve random splitting — the largest such effect documented in UAV crop classification literature, with direct implications for the validity of published benchmarks. (2) Classical HOG models (SVM macro-F1=0.388, RF=0.392) fail completely on bicycle, motor, and people minority classes despite balanced training, demonstrating that hand-crafted descriptors are insufficient for the full distributional complexity of this class set. (3) ResNet18 with Focal Loss (macro-F1=0.497) is the only model achieving non-zero F1 for four minority classes (bicycle: 0.538, motor: 0.719, people: 0.463), with a Top-2 accuracy of 96.9% confirming strong soft discriminability. (4) The t-SNE output directly confirms bus and bicycle scarcity as the mechanistic cause of zero F1 — not model inadequacy — providing interpretable, causal evidence that should be considered a standard component of classification evaluation reporting. Multi-sequence training and full-frame detection architectures are the recommended next steps.

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